CBSE Class 12 Mathematics Code 041 MARKING SCH

Annual Examination 2023 - Solution & Marking Breakdown

Q1: |adj A| = |A|^n - 1 = |A|^2 = 64 => |A| = +/- 8 [1 mark]

Q2: Reflexive: a - a = 0 divisible by 2 [0.5 mark]
Symmetric: a - b = 2k => b - a = -2k [0.5 mark]
Transitive: a - b + b - c = 2k1 + k2 [1 mark]

Q3: Let I = int_0^pi/2 sin^4 x / sin^4 x + cos^4 x dx
Use property I = int_0^pi/2 cos^4 x / cos^4 x + sin^4 x dx [1.5 marks]
Adding both: 2I = int_0^pi/2 1 dx = pi/2 => I = pi/4 [1.5 marks]